

SHAMIK BASU

Los Angeles, CA | (213) 255-6219 | shamik1900@gmail.com | linkedin.com/in/shamikofficial | github.com/ShamikOfficial | shamik-basu.com

SUMMARY

Data scientist deploying AI and machine learning pilot solutions in Python across data ingestion, model deployment, and REST API integration. Career highlights: document processing and LLM applications in regulated healthcare on 5M+ records/month (92% accuracy, -72% cost), rapid prototyping in Agile delivery, and decision-support analytics with written and verbal communication skills for stakeholders.

EDUCATION

University of Southern California | Masters, Applied Data Science January 2025 - December 2026

Relevant Coursework: Foundations of AI, Machine Learning, NLP, Statistical Methods, Data Mining

SRM Institute of Science & Technology | Bachelor, Computer Science May 2018 - June 2022

Relevant Coursework: Data Structures, Algorithms, Operating Systems, Advanced C/C++

WORK EXPERIENCE

Data Science Associate Intern (Part-Time) | Liquidity AI Capital | Los Angeles, CA January 2026 - May 2026

- Built Python LLM pilot with LoRA fine-tuning, data ingestion, and model evaluation on 12K+ financial documents.
- Developed agentic LangChain prototype with REST API integration and entity classification at 88% accuracy.
- Deployed Docker inference with documentation on data flows, model behavior, and debugging for sustainment.

Data Scientist | Bajaj Finserv Health | Pune, India November 2023 - December 2024

- Deployed AI/ML document processing (LangChain + GPT-3.5 Turbo + REST APIs) on 5M+ records/month at 92% accuracy with -72% cost in regulated healthcare.
- Added Langfuse MLOps observability for model lifecycle management, cutting GPU waste 15% across inference pipelines.
- Integrated Apache Kafka and APIs for end-to-end analytical pipelines from pilot to scalable production use.

Associate Data Scientist | Bajaj Finserv Health | Pune, India July 2022 - October 2023

- Built predictive models and lead scoring with scikit-learn for Salesforce decision-support campaigns.
- Improved natural language processing extraction 27% by redesigning NER pipeline on inconsistent data.
- Delivered workflow automation gains of 22% with logistic regression and verbal communication skills in stakeholder reporting.

Data Engineer Intern | Bajaj Finserv Health | Pune, India January 2022 - June 2022

- Built Python and C++ pipelines ingesting and transforming 10M+ records across 200+ features on an enterprise cluster.
- Integrated REST API-connected ETL with Trino and Docker, cleaning operational datasets for downstream modeling.
- Documented data flows and debugged quality issues blocking production ML workflows in a regulated system.

SKILLS

Programming Languages: Python, SQL, C, C++

AI & ML Solutions: Artificial intelligence, machine learning, natural language processing, LLM applications, LangChain, RAG, predictive models

Pipelines & Deployment: Data ingestion, data processing, model deployment, MLOps, REST APIs, workflow automation, Docker, Kafka

Delivery & Cloud: Agile prototyping, rapid prototyping, Microsoft Azure, Amazon Web Services, technical documentation

PROJECTS

EcoMateAI | UCLA SAIRS Hackathon Winner May 2025 - June 2025

GitHub: github.com/ShamikOfficial/EcoMate-AI | **Live:** ecomateai.streamlit.app

- Shipped Agile AI pilot (Gemini 2.5 Flash) with document processing, OCR, and multimodal data ingestion.
- Deployed Streamlit prototype with API integrations, JSON parsing, and technical docs for user testing.

Agentic Trend Orchestrator May 2026 - June 2026

GitHub: github.com/ShamikOfficial/agentic-trend-orchestrator | **Live:** trendpilot.shamik-basu.com

- Built LLM application prototype (FastAPI + Docker) with RAG retrieval, API integration, and iterative Agile delivery.
- Documented model behavior and eval logging for prototype sustainment and knowledge transfer.

Custom CUDA Library January 2026 - February 2026

GitHub: github.com/ShamikOfficial/CUDA-Accelerated-Python-Library | **Technical blog:** shamik-basu.com/projects/custom-cuda-library.html

- Accelerated Python data processing with C/C++ CUDA kernels; 115x CPU speedup on matrix workloads.
- Published technical documentation with reproducible benchmarks for preprocessing workflows.

LEADERSHIP

- VP of GRIDS, a 250-member USC club running hackathons, workshops, and technical talks.
- Graduate Mentor, USC Viterbi School of Engineering.
- Volunteer, USC Trojan Pantry, distributing food to students facing financial need.